

Gerrymandering Analysis of Massachusetts Congressional Districts

CS464 — AI for Redistricting | Spring 2026
University of San Francisco

2024 Presidential & Senate Elections | ReCom Ensemble (40,000 steps)

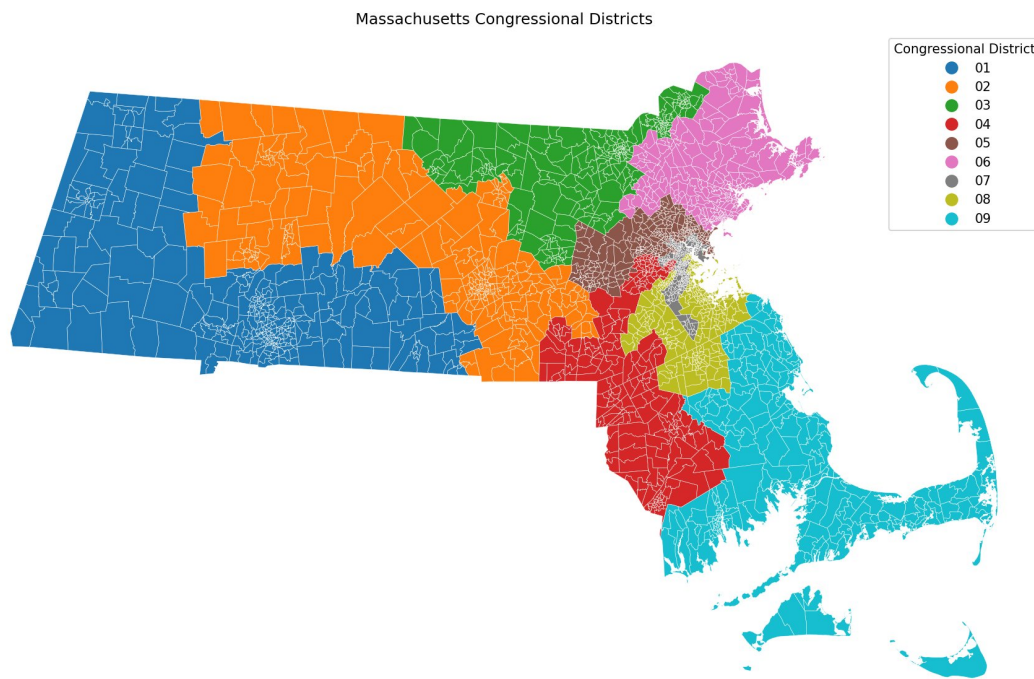


Figure 1. The enacted Massachusetts congressional district map. Western districts are geographically large; Boston-area districts are small but densely populated.

1. Introduction

This report uses ensemble analysis to evaluate the enacted Massachusetts congressional map. We generated a large collection of alternative Massachusetts maps — all meeting the same legal requirements — and asked whether the enacted map's outcomes stand out as unusual compared to that collection. If they do, that is evidence the map was shaped to achieve those outcomes rather than simply reflecting where people live.

The central challenge with Massachusetts is that it is a deeply blue state, so Democratic-favoring outcomes are expected under almost any valid map. The analysis has to distinguish between outcomes that are unusual relative to the ensemble versus outcomes that simply reflect the state's partisan geography.

All code and data for this project are available at:
<https://github.com/WeeWolfG/CS464-final-project>

2. State and Map Context

Political landscape.

Massachusetts is a deeply blue state. Kamala Harris won it by roughly 25 points in 2024, and Democrats hold all nine congressional seats. This matters a lot for interpreting the analysis: Democratic-favoring outcomes are expected under almost any reasonable Massachusetts map. The real question is not whether the enacted map helps Democrats — it clearly does — but whether it helps them *more* than other valid Massachusetts maps would.

Map creation and legal context.

The current map was enacted in 2021. Democratic-controlled state lawmakers drew it through the normal legislative process — the governor could veto it, and a two-thirds majority could override that veto. Massachusetts has no independent redistricting commission. The map has not faced any significant legal challenges.

Racial landscape and District 7.

Massachusetts is about 71% white, with Black, Hispanic, and Asian communities concentrated in greater Boston. District 7 — covering Boston, Cambridge, and Somerville — is the state's only majority-minority district, at 61.33% minority population according to the 2021 mapbook. In our precinct data it shows an 82.6% Democratic two-party vote share for President and 82.3% for the Senate race, making it the state's bluest district by a wide margin. This creates a real interpretive tension: some of that Democratic concentration may be legally required to preserve minority voting power under the Voting Rights Act, not purely a product of partisan line-drawing.

3. Data

Three source files were used:

File	Description	Rows
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ma_pl2020_p1_b.shp	2020 Census population at the block level. Column P0010001 gives total population per block.	107,278
ma_pl2020_cnty.shp	Massachusetts county boundaries. Used as nesting regions during precinct geometry repair.	14
ma_2024_gen_all_prec.shp	2024 general election precincts with vote totals for the Presidential and Senate races, plus congressional district assignment (CONG_DIST).	2,383

The two elections are the 2024 Presidential race (Harris vs. Trump; columns G24PREDHAR and G24PRERTRU) and the 2024 U.S. Senate race (Warren vs. Deaton; columns G24USSDWAR and G24USSRDEA). Running the analysis on two independent races lets us check whether the findings hold across different electoral contexts or are specific to one.

4. Methodology

4.1 MAUP Data Cleaning Pipeline

Before any analysis could run, the three shapefiles had to be cleaned and aligned. This used the MAUP library and followed five steps.

- **Reprojection.** All three files were reprojected to the same UTM coordinate system (Zone 19N), estimated from the election precinct file. This ensures distances and areas are measured in meters rather than degrees.
- **Geometry repair.** The raw precinct file had 519 overlapping areas, 77 gaps, and invalid geometries — normal for government shapefiles. We applied `smart_repair` with `nest_within_regions=county_df`, which fixed all overlaps and holes while keeping precincts inside county lines. `snap_precision=10` was tried first but failed for this dataset, so `snap_precision=8` was used as a fallback.
- **Rook-to-queen conversion.** Any two precincts sharing a border shorter than 30 meters (`min_rook_length=30`) were converted from a shared edge to a point contact. This prevents tiny artifact slivers from creating spurious neighbor relationships in the GerryChain graph.
- **Population assignment.** `maup.assign` mapped each of the 107,278 census blocks to its containing precinct, and block populations were summed up to produce a TOTPOP value per precinct. A hard assertion confirmed that the block total equals the precinct total: both are 7,029,917, so no population was lost.
- **Validation.** `maup.doctor` returned True on the cleaned file. The shapefile was saved to MA/MA.shp and a config.json file recorded the column names and parameter choices for the analysis notebook.

4.2 Graph Construction and Island Patching

GerryChain works with a graph, not a shapefile: each precinct is a node, and two precincts are connected by an edge if they share a border (queen adjacency). We built this graph from the

cleaned shapefile using `Graph.from_file`, loading population, district assignment, and all four vote columns onto each node.

The disconnected component problem. Massachusetts has offshore islands — most notably Martha's Vineyard and Nantucket, but also several smaller islands off the Boston Harbor and South Shore coastline — that are assigned to a congressional district but share no physical border with the mainland portion of that district. In graph terms, these islands form disconnected components: they are valid nodes but have no edges connecting them to the rest of their district. GerryChain's contiguity constraint requires every district in every proposed map to be a single connected component. Without patching, the chain would reject every single proposal because the starting partition would itself violate contiguity, making it impossible to run at all.

Our approach: centroid distance minimization with manual verification. For each disconnected district, we identified all connected components, then found the pair of precincts — one from the main body, one from the isolated piece — with the shortest straight-line distance between their geographic centroids. We added a single edge between that pair in the GerryChain graph. This edge exists only in the graph data structure and does not alter any precinct boundary. After all edges were added, we verified each connection visually using dual graphs: zoomed plots showing the two connected precincts, the red edge between them, and their surrounding neighbors. This step lets us confirm that each connection makes geographic sense — for example, that an island precinct is being connected to the nearest feasible mainland precinct rather than an unrelated one on the other side of a bay. A total of seven edges were added across districts 07, 08, and 09.

Why this approach, and what are the trade-offs? Centroid distance is fast and deterministic: for a graph with 2,383 nodes it runs in seconds. The main alternative — finding the closest pair of boundary points rather than centroids — would require comparing thousands of coordinate pairs per precinct and is computationally expensive at this graph size without meaningful benefit, since the goal is simply to assert graph adjacency rather than to place a physical boundary. Another alternative — snapping each island to the nearest precinct regardless of district — risks creating cross-district edges that confuse the partition logic. The centroid-within-district approach avoids this by only searching for partner precincts inside the same district. The main limitation of centroid distance is that for very irregular precinct shapes, the centroid can lie outside the precinct polygon, so the shortest centroid-to-centroid pair may not correspond to the geographically closest boundary pair. The dual graph visualizations below let us catch and correct any such cases manually.

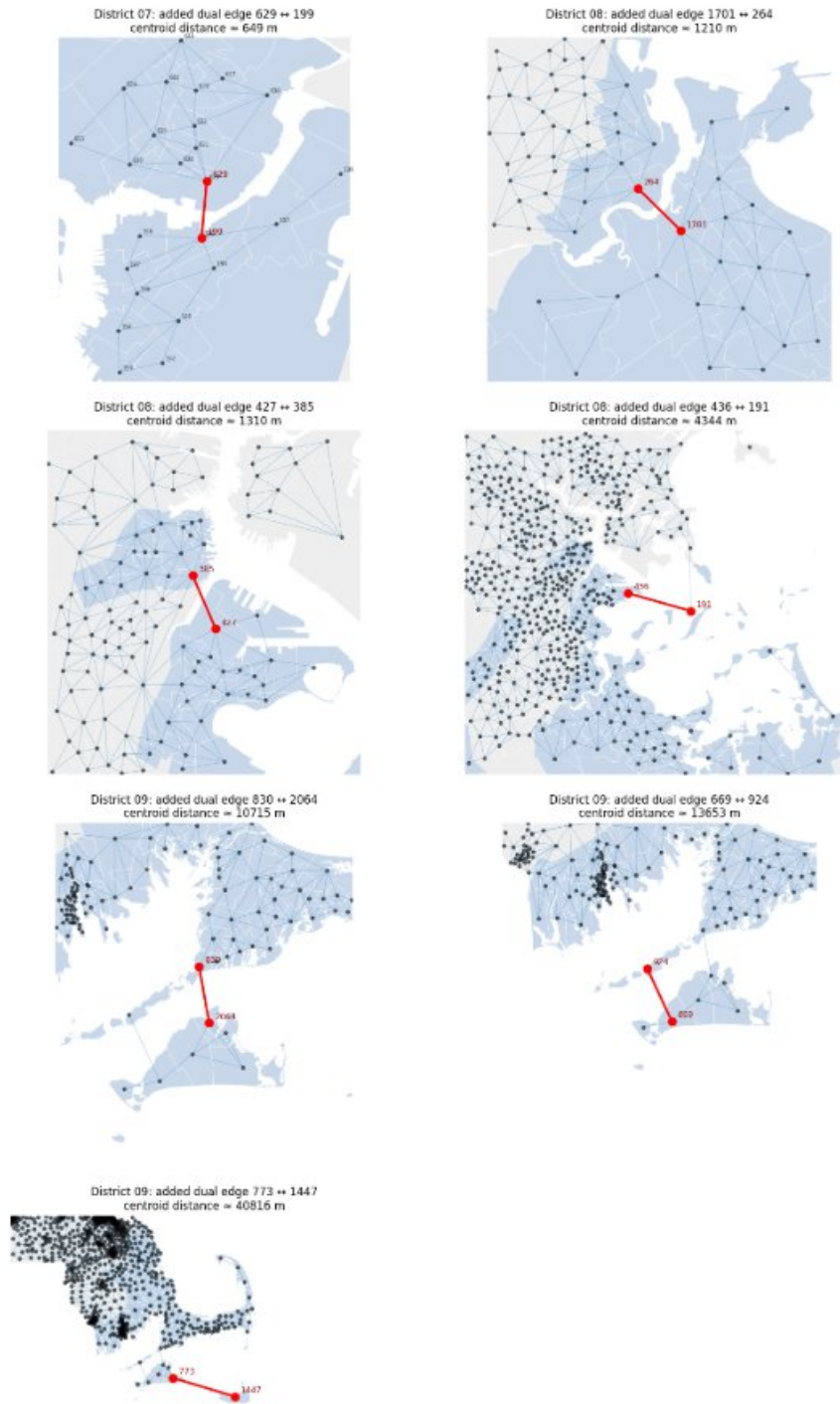


Figure 2. Dual graph visualizations of the seven manually added island connections across districts 07, 08, and 09. Each subplot shows the two connected precincts (red dots), the added edge (red line), and their geographic context. Centroid distances range from 649 m (District 07) to 40,816 m (District 09, Nantucket connection).

4.3 ReCom Ensemble Method

ReCom (Regional Recombination) generates alternative maps by repeatedly taking one random step: two neighboring districts are merged into one region, a random spanning tree is

built over that region, and one edge of the tree is cut to produce two new valid districts. Because the spanning tree is built randomly each time, each step produces a genuinely different map rather than a small tweak of the previous one.

The following parameters were used:

Parameter	Value	Justification
Population tolerance	$\pm 2\%$	Standard class setup; ensures legal equal-population compliance.
Node repeats	2	Recommended by GerryChain documentation for graphs of this size.
Compactness bound	$\leq 2 \times$ initial cut edges	Stops the chain from exploring wildly fragmented maps while still letting it range above and below the enacted map's compactness.
Chain length	20,000 and 40,000 steps	Both lengths were run. The 40k ensemble is used for all final figures; convergence confirms 20k was already sufficient.
Elections	PRES24 and SEN24	Two independent elections; findings consistent across both.

At every step we recorded: cut edges, Democrat-won districts, efficiency gap, and sorted Democratic vote share per district (for the marginal box plots). The same values were computed for the enacted map to serve as reference points.

4.4 Metrics

Efficiency Gap. The efficiency gap measures partisan asymmetry through wasted votes. A vote is wasted if it goes to the losing party (all of those are wasted) or if it goes to the winner beyond what was needed to win ($50\% + 1$ vote). The gap is the difference in total wasted votes between the two parties, divided by total votes cast. A negative value means more Democratic votes were wasted, so the map favors Democrats. A common benchmark is that values beyond $\pm 8\%$ are worth flagging.

Cut edges. A cut edge is a border between two precincts that land in different congressional districts. More cut edges means more fragmented, convoluted boundaries. Comparing the enacted map's count to the ensemble distribution shows whether its boundaries are unusually simple or unusually messy.

Marginal box plots. At each step, the nine districts are ranked by two-party Democratic vote share from lowest to highest. Box plots show the distribution of each rank's vote share across the entire ensemble, with the enacted map's values overlaid as dots. This visualization —

sometimes called the signature of gerrymandering — reveals whether the enacted map's individual districts are unusually safe or competitive compared to what the ensemble typically produces.

5. Results

5.1 Convergence

Before trusting the ensemble results, we need to verify the chain ran long enough to genuinely explore the space of valid maps. Figure 2 shows the running average of cut edges over all steps for both the 20k and 40k chains. Both drop sharply from the starting value of 659 — the enacted map's count — and settle near 486 within the first few thousand steps. The two chains land at the same overall average, and both traces remain flat for the rest of the run. The chain has converged, and doubling the length doesn't change the story.

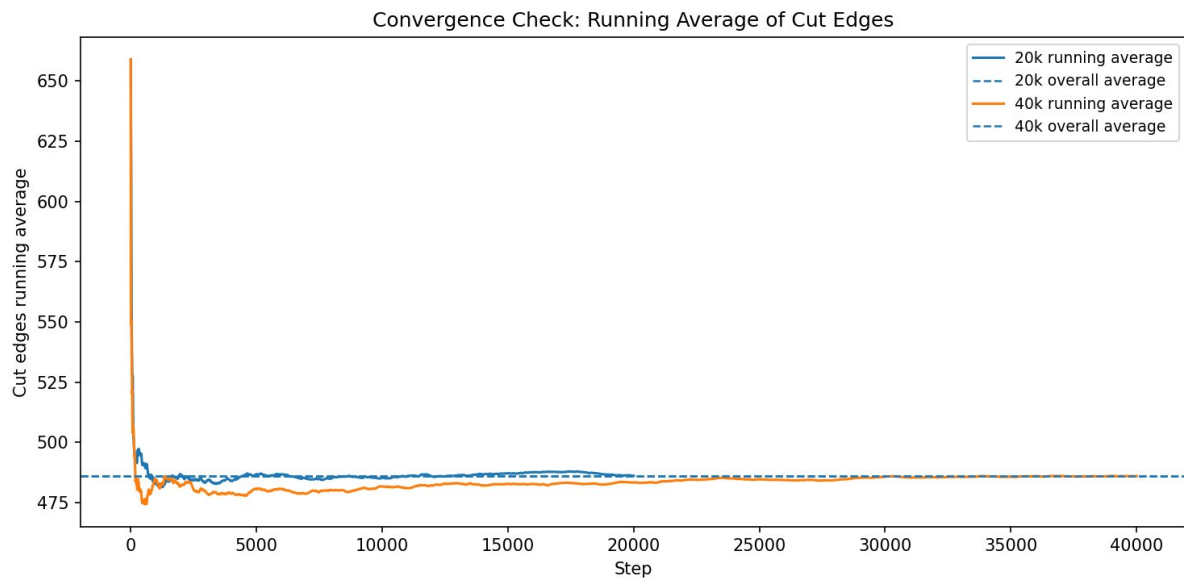


Figure 3. Running average of cut edges for the 20k and 40k chains. Both stabilize near 486 within a few thousand steps. Dashed lines show each chain's overall mean.

Figure 4 overlays the 20k and 40k distributions for all three metrics and both elections. The histograms overlap almost perfectly in every panel, confirming that the 40k run adds confidence without changing the results. All final figures use the 40k ensemble.

Convergence Evidence: 20k vs 40k Ensembles

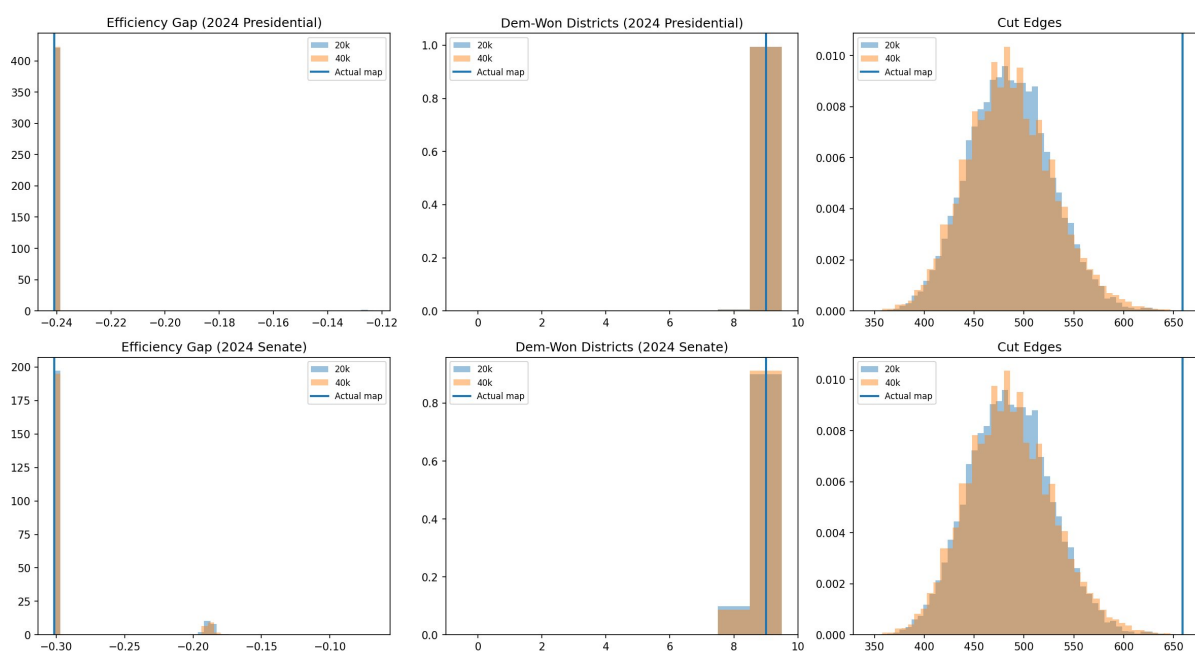


Figure 4. 20k and 40k ensemble distributions overlaid for efficiency gap, Democrat-won districts, and cut edges. The strong overlap confirms convergence. The blue vertical line marks the enacted map's value.

5.2 Enacted Map Metrics

Key metric values for the enacted Massachusetts map:

Metric	2024 Presidential	2024 Senate
Democratic wins	9 / 9	9 / 9
Efficiency Gap	-0.2409	-0.3015
Ensemble mean EG (40k)	-0.2403	-0.2915
Cut edges	659	659 (same graph)
Ensemble mean cut edges (40k)	~486	~486

5.3 Outlier Analysis

Figure 5 shows the full 40k ensemble distributions for all three metrics.

40k ReCom Ensemble Outlier Analysis

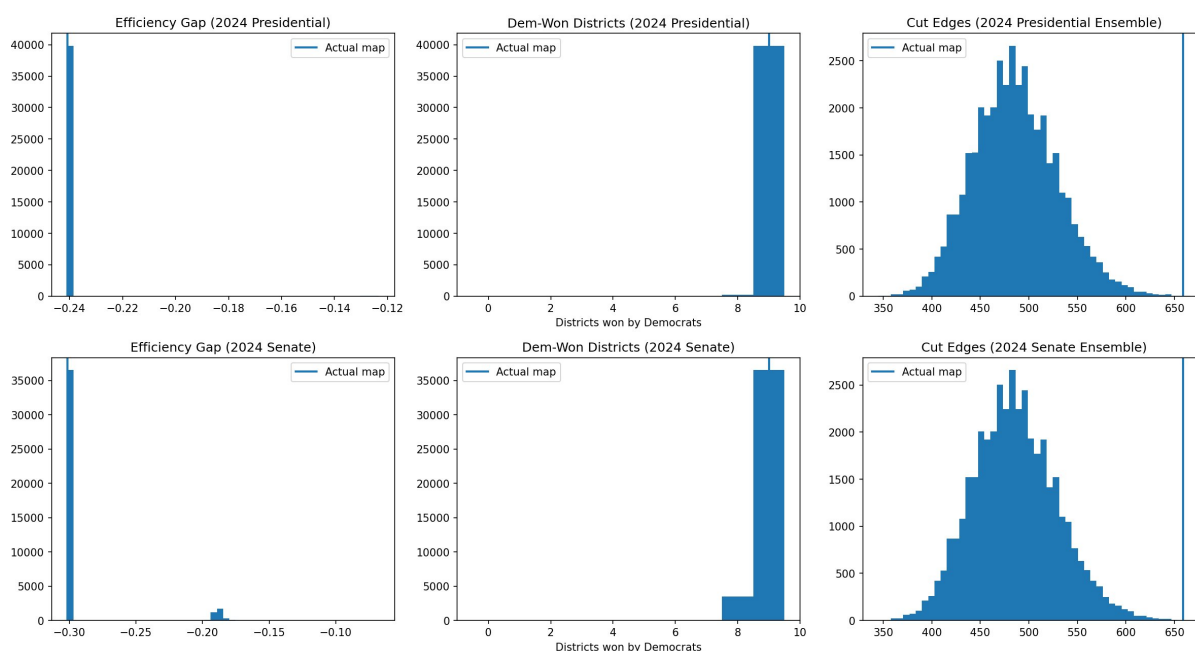


Figure 5. 40k ReCom ensemble outlier analysis. Each histogram shows a metric's distribution across 40,000 generated maps; the blue line marks the enacted map. Top row: 2024 Presidential. Bottom row: 2024 Senate.

Efficiency Gap. Both EG histograms form a narrow spike: near -0.24 for the Presidential race and near -0.20 for the Senate. The enacted map's Presidential EG of -0.2409 lands right on the ensemble mean of -0.2403 — it is not an outlier at all. The Senate EG of -0.3015 is a bit further left than the ensemble mean of -0.2915, but not dramatically so. Massachusetts is so uniformly Democratic that almost any valid map produces a strongly negative EG, so this metric doesn't give us much to work with.

Democrat-won districts. For the Presidential race, the ensemble gives Democrats 9 wins in nearly every single map — the enacted map's 9-for-9 result is the most common outcome, not an outlier. For the Senate race, about 9% of ensemble maps produce only 8 Democratic wins, placing the enacted map's 9-for-9 at the right edge of the distribution. This reflects the state's partisanship: in a state this blue, sweeping all nine seats is essentially expected for Presidential races and common for Senate ones.

Cut edges. This is where the clearest signal shows up. The ensemble is centered near 486 cut edges, forming a bell curve from roughly 380 to 580. The enacted map's value of 659 sits entirely to the right of that distribution — beyond the visible range of the histogram. The enacted boundaries are substantially more fragmented than virtually any map ReCom produces under the same constraints, and this holds for both elections since cut edges depend only on the district geometry.

5.4 Marginal Box Plots

Figure 6 shows the marginal box plots. Each x-axis position is a rank: rank 1 is the least Democratic district in a given map, rank 9 is the most Democratic. This is a ranking within each

individual map, not a label for a specific congressional district number. The box shows the ensemble distribution at that rank, and the blue dots show the enacted map. The dashed line at 0.5 is the majority threshold.

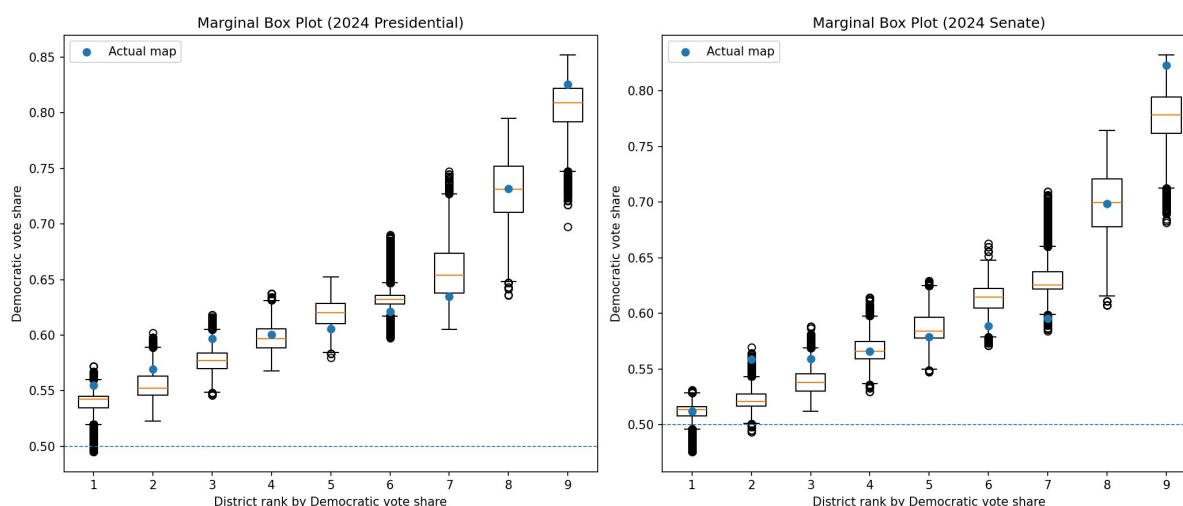


Figure 6. Marginal box plots using the 40k ensemble. Each box shows the distribution of Democratic vote share at that rank; blue dots show the enacted map. Left: 2024 Presidential. Right: 2024 Senate.

All nine enacted districts sit above 50% in both elections, consistent with Democrats winning every seat. For the Presidential race, the enacted map's dots track close to the ensemble medians at ranks 1 through 7. The clearest deviation is at rank 9, where the enacted map's bluest district (around 82% Democratic) sits noticeably above the ensemble median — consistent with the heavy Democratic concentration in District 7.

The Senate race shows more scatter. Several enacted-map districts fall outside the ensemble boxes at middle ranks, and the bluest district again sits above the median at rank 9. The middle-rank scatter doesn't have a single clean explanation — it may reflect how the enacted boundaries interact differently with the Senate vote distribution than the Presidential one, or small vote-share differences between the two races in competitive precincts.

5.5 Compactness: Polsby-Popper

As a supplementary check, we computed Polsby-Popper scores for each district in the enacted map. The formula is $4\pi \times \text{Area} / \text{Perimeter}^2$ — a perfect circle scores 1.0, and more jagged or irregular shapes score closer to 0.

District	Polsby-Popper	District	Polsby-Popper
01	0.2797	06	0.2654
02	0.2119	07	0.1823
03	0.3264	08	0.1934
04	0.2341	09	0.2512

05	0.3876	Mean	0.2702
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Districts 7 and 8 have the lowest scores (0.1823 and 0.1934), which makes sense given their irregular shapes in the dense Boston area. District 5 is the most compact at 0.3876. All scores are well below 1.0, consistent with the high cut edge count: fragmented district boundaries naturally produce low per-district compactness scores.

6. Discussion

The results are a mixed picture, and Massachusetts's political context is essential for reading them correctly.

What the analysis actually finds. The strongest signal is boundary fragmentation. At 659 cut edges against an ensemble mean of 486, the enacted map sits well outside the distribution of randomly generated alternatives. This holds across both chain lengths and both elections, making it the most robust finding. By contrast, the efficiency gap is not a strong signal — the enacted map's Presidential EG of -0.2409 is essentially the same as the ensemble mean of -0.2403. The Senate EG shows a bit more separation, but not dramatically so. Winning all nine seats is the expected outcome under any reasonable Massachusetts map for a Presidential race and common for a Senate race. These metrics do not flag the enacted map as unusual.

Geography versus intent. A state with heavy Democratic concentration in urban areas will naturally produce some Democratic-favoring outcomes regardless of how lines are drawn. But the cut edge excess — 659 versus 486 — is hard to explain away purely by geography. The compactness bound we imposed allows the chain to produce maps up to twice the enacted map's cut edges, yet it almost never gets anywhere near 659. That suggests the enacted map's boundary complexity goes beyond what Massachusetts's population distribution requires.

The role of District 7. District 7 complicates the picture because it serves two functions at once. As the state's only majority-minority district, it protects the political representation of Boston-area communities of color and may be legally required to maintain that majority under the Voting Rights Act. At the same time, it is the state's bluest district by a wide margin — 82.6% Democratic for President — and it drives the packing signal at rank 9 in the marginal box plots. These two roles can't be cleanly separated without additional ecological inference analysis: some of that Democratic concentration may be VRA-required, while some may reflect additional partisan choices about exactly which precincts were included.

Limitations. The ensemble cannot explore maps more fragmented than twice the enacted map's cut edges, so our comparison set is bounded. The analysis also doesn't account for criteria like communities of interest or political subdivision boundaries. The Polsby-Popper scores are computed from precinct-level geometries rather than smooth district outlines and may be somewhat noisy. Finally, the `snap_precision=8` fallback during geometry repair, while producing a passing `maup.doctor` result, is slightly less strict than the default of 10; any practical effect on the analysis is expected to be small.

7. Conclusion

The ensemble analysis turns up one clear outlier and several expected results. The enacted map has 659 cut edges while 40,000 randomly generated alternatives consistently land near 486. That degree of boundary fragmentation places the enacted map well outside what neutral map-drawing produces, and it is the strongest evidence that the district lines were deliberately shaped rather than following the natural geography of the state.

The efficiency gap and seat count findings are more nuanced. The enacted map's Presidential EG of -0.2409 is essentially identical to the ensemble mean of -0.2403, meaning it doesn't produce Democratic vote-wasting advantages beyond what any randomly drawn Massachusetts map would. Democrats winning all nine seats is the expected outcome in a state this blue. These metrics don't flag the enacted map as unusual.

The marginal box plots confirm Democratic dominance across all districts and show that the enacted map's bluest district — District 7, the Boston-area majority-minority seat — is more Democratic than the ensemble typically produces at that rank. Separating how much of that is legally required by the VRA from how much reflects extra partisan packing would require ecological inference analysis beyond the scope of this project.

In short, Massachusetts has a map with unusually fragmented boundaries that stand out clearly relative to the ensemble. Whether that fragmentation reflects deliberate partisan manipulation, the genuine complexity of maintaining minority representation in a dense urban geography, or some combination of both is the central question this analysis raises — and cannot fully resolve.

References

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